

# AtmoVista - Internship Project Report

End-to-end AQI forecasting for Pakistan with MLOps, live deployment, and GitHub Actions automation.

**Project title:** AtmoVista (Pearls AQI Predictor)

**Forecast window:** Next 3 days (+24h, +48h, +72h)

**Cities:** Lahore, Karachi, Islamabad, Peshawar, Quetta

**Repository:** Hosted on GitHub with GitHub Actions CI/CD

**Deployment:** API on Render, frontend on Vercel, features/models on Hopsworks

## 1. Introduction

AtmoVista is an end-to-end machine learning system that predicts Air Quality Index (AQI) for five major Pakistan cities over the next three days. The full pipeline is implemented, pushed to GitHub, automated with GitHub Actions, and deployed for live use.

The system includes:

- automated data collection
- feature engineering and historical backfill
- multi-model training and evaluation
- Hopsworks feature store and model registry
- CI/CD with GitHub Actions (hourly features + daily training)
- interactive AtmoVista web dashboard (FastAPI + React)

The goal is practical: help everyday users understand current and upcoming air quality, while also showing model performance and explainability for technical review.

## 2. Problem statement

Air quality in Pakistan varies strongly by city and season. A useful system should:

- Forecast AQI for the next 24, 48, and 72 hours
- Update regularly with fresh data
- Explain predictions (not just show a number)
- Alert when conditions become unhealthy or hazardous
- Run on a free / serverless-friendly stack

AtmoVista addresses these needs with an MLOps pipeline and a dual-mode dashboard (For you / For experts).

## 3. Objectives completed

| Objective                                 | Status                |
|-------------------------------------------|-----------------------|
| Predict AQI for next 3 days               | Completed             |
| Feature pipeline (fetch to feature store) | Completed             |
| Historical backfill for training          | Completed             |
| Train and evaluate multiple models        | Completed             |
| Store models in model registry            | Completed             |
| Automate hourly and daily pipelines       | Completed and running |
| Interactive AtmoVista dashboard           | Completed             |

| Objective                            | Status    |
|--------------------------------------|-----------|
| EDA                                  | Completed |
| SHAP explainability                  | Completed |
| Hazardous AQI alerts                 | Completed |
| GitHub + Render + Vercel + Hopsworks | Completed |

UI note: The brief mentions Streamlit/Gradio. This project uses FastAPI + React because the mentor allowed a stronger custom frontend for higher UI marks.

## 4. System overview

AtmoVista follows this flow in production:

- Open-Meteo APIs provide pollutant and weather data
- Feature and backfill pipelines engineer hourly features
- Features are stored in the Hopsworks Feature Store
- Training compares Ridge, Random Forest, HistGradientBoosting, and LSTM
- Winning models are registered in the Hopsworks Model Registry
- FastAPI loads models/features and serves forecasts
- The AtmoVista React dashboard consumes the API
- GitHub Actions keeps features and models updated on schedule

### Main layers

| Layer        | Responsibility                                          |
|--------------|---------------------------------------------------------|
| Data         | Open-Meteo air quality and weather                      |
| Features     | Time, lag, rolling, future-weather features and targets |
| Storage      | Hopsworks Feature Group and Model Registry              |
| Training     | Per-horizon model selection                             |
| Serving      | FastAPI inference API                                   |
| Presentation | AtmoVista React dashboard                               |
| Automation   | GitHub Actions hourly and daily jobs                    |

## 5. Technology stack

| Area          | Choice          | Reason                         |
|---------------|-----------------|--------------------------------|
| Language      | Python          | Required for ML pipelines      |
| Classical ML  | Scikit-learn    | Ridge / RF / HistGB            |
| Deep learning | TensorFlow LSTM | Advanced model requirement     |
| Feature store | Hopsworks       | Feature Store + Model Registry |

| Area           | Choice                    | Reason                            |
|----------------|---------------------------|-----------------------------------|
| Orchestration  | GitHub Actions            | Free CI/CD alternative to Airflow |
| API            | FastAPI                   | Backend for serving forecasts     |
| UI             | React + Vite + TypeScript | Mentor-approved custom frontend   |
| Explainability | SHAP                      | Required XAI                      |
| Deploy         | Render + Vercel           | Free-tier production hosting      |

## 6. Data collection

Source: Open-Meteo Air Quality API and Weather API (free, no paid key). The brief lists AQICN/OpenWeather as examples and allows exploring other options.

### Pollutants / AQI

- PM2.5, PM10, CO, NO2, SO2, O3
- US AQI (preferred) / European AQI fallback

### Weather

- temperature, humidity, precipitation
- wind speed / direction
- cloud cover, surface pressure

### Cities and coordinates

| City      | Latitude | Longitude |
|-----------|----------|-----------|
| Lahore    | 31.5204  | 74.3587   |
| Karachi   | 24.8607  | 67.0011   |
| Islamabad | 33.6844  | 73.0479   |
| Peshawar  | 34.0151  | 71.5249   |
| Quetta    | 30.1798  | 66.9750   |

### Training scale

| Item               | Value              |
|--------------------|--------------------|
| Backfill window    | 365 days           |
| Dataset size (EDA) | 43,800 hourly rows |
| Cities covered     | 5                  |

## 7. Feature engineering

Implemented in `src/features/engineering.py`.

| Group            | Examples                                            |
|------------------|-----------------------------------------------------|
| Raw signals      | pollutants, weather, current AQI                    |
| Time features    | hour, day, day_of_week, month, sin/cos encodings    |
| Derived features | AQI change rate, temp x humidity, wind x PM2.5      |
| Lag features     | AQI and PM2.5 lags at 1, 3, 6, 12, 24, 48, 72 hours |
| Rolling stats    | mean/std over 6h, 12h, 24h windows                  |
| Future weather   | weather at t+24 / t+48 / t+72                       |
| Targets          | aqi_target_24h, aqi_target_48h, aqi_target_72h      |

Design choices:

- Per-horizon models instead of one multi-output model (better accuracy)
- Realistic mode adds controlled noise to future-weather features during training
- Prediction intervals (80%) estimated from validation residuals

## 8. Exploratory data analysis (EDA)

EDA notebook: notebooks/eda.ipynb. Plots include timeseries, correlation heatmap, seasonality, and city boxplots.

Key findings:

- Lahore has the highest median AQI and widest spread (strong smog spikes).
- Karachi is the most stable city with the lowest mean AQI in this dataset.
- PM2.5 and PM10 are the strongest linear correlates of AQI.
- Evening/night hours and winter months (Nov to Jan) tend toward higher mean AQI.
- City-specific behavior justifies shared features but different error profiles.

### City AQI summary (from EDA)

| City      | Mean  | Std  | Median | Max |
|-----------|-------|------|--------|-----|
| Islamabad | 113.0 | 30.7 | 107.0  | 206 |
| Karachi   | 90.1  | 21.8 | 85.0   | 165 |
| Lahore    | 153.0 | 49.1 | 154.0  | 364 |
| Peshawar  | 114.6 | 29.0 | 111.0  | 212 |
| Quetta    | 86.4  | 31.7 | 77.0   | 332 |

### Forecast difficulty by city

| Difficulty  | Cities            |
|-------------|-------------------|
| Hardest     | Lahore, Quetta    |
| More stable | Karachi, Peshawar |

## 9. Training pipeline

Script: pipelines/training\_pipeline.py

### Models experimented

| Model                | Library      |
|----------------------|--------------|
| Ridge Regression     | Scikit-learn |
| Random Forest        | Scikit-learn |
| HistGradientBoosting | Scikit-learn |
| LSTM                 | TensorFlow   |

### Evaluation metrics

| Metric            | Purpose                                |
|-------------------|----------------------------------------|
| RMSE              | Main selection metric                  |
| MAE               | Average absolute error                 |
| R2                | Explained variance                     |
| Category accuracy | AQI band match for user-facing quality |

Best model is selected per horizon using validation RMSE.

| Horizon | Selected model       |
|---------|----------------------|
| +24h    | Ridge                |
| +48h    | HistGradientBoosting |
| +72h    | Ridge                |

Winner label: ridge@24h+hist\_gradient\_boosting@48h+ridge@72h | Trained: 2026-08-11

## 10. Results

### 10.1 Validation metrics (winner models)

| Horizon | Winner | RMSE  | MAE   | R2    | Category Acc. |
|---------|--------|-------|-------|-------|---------------|
| +24h    | Ridge  | 21.63 | 14.33 | 0.716 | 72.8%         |
| +48h    | HistGB | 26.22 | 18.64 | 0.583 | 59.9%         |
| +72h    | Ridge  | 28.47 | 19.78 | 0.508 | 61.2%         |

### 10.2 Test metrics (winner models)

| Horizon | Winner | RMSE  | MAE   | R2    | Category Acc. |
|---------|--------|-------|-------|-------|---------------|
| +24h    | Ridge  | 20.63 | 14.77 | 0.611 | 67.8%         |

| Horizon | Winner | RMSE  | MAE   | R2    | Category Acc. |
|---------|--------|-------|-------|-------|---------------|
| +48h    | HistGB | 26.63 | 19.26 | 0.352 | 58.1%         |
| +72h    | Ridge  | 28.18 | 20.90 | 0.275 | 54.9%         |

### 10.3 Interpretation

- Short-horizon (+24h) forecasts are clearly strongest.
- Error grows with horizon, which is expected for air-quality forecasting.
- Models beat a simple persistence baseline on the tracked horizons.
- Category accuracy matters because users act on bands, not only exact AQI.

### 10.4 Live monitoring (forecast vs actual)

Source: artifacts/monitoring/summary.json

| Scope   | MAE   | RMSE  | R2    | Category Acc. | Scored rows |
|---------|-------|-------|-------|---------------|-------------|
| Overall | 12.33 | 19.32 | 0.744 | 74.8%         | 3500        |
| +24h    | 10.42 | 15.84 | 0.830 | 78.8%         | 1170        |
| +48h    | 13.48 | 20.56 | 0.709 | 72.1%         | 1165        |
| +72h    | 13.10 | 21.13 | 0.691 | 73.4%         | 1165        |

### Per-city live MAE

| City      | MAE   | Category Acc. | Scored rows |
|-----------|-------|---------------|-------------|
| Karachi   | 4.68  | 97.6%         | 700         |
| Islamabad | 13.15 | 66.7%         | 700         |
| Peshawar  | 12.95 | 70.1%         | 700         |
| Quetta    | 14.51 | 71.1%         | 700         |
| Lahore    | 16.36 | 68.3%         | 700         |

Karachi performs best in live monitoring. Lahore is the hardest city among the five.

## 11. Feature store and model registry (Hopsworks)

| Item             | Value                                             |
|------------------|---------------------------------------------------|
| Feature Group    | aqi_features (v3)                                 |
| Feature contents | Engineered hourly features + targets for 5 cities |
| Model Registry   | aqi_forecast_model                                |
| Bundle           | Per-horizon winners synced by hopsworks_sync.py   |

Production mode (STORAGE\_MODE=hopsworks) loads features from Hopsworks and keeps an in-memory cache in the API. The deployed app uses Hopsworks as the source of truth.

## 12. Inference and API

API module: api/main.py

| Endpoint area                  | What it serves                                |
|--------------------------------|-----------------------------------------------|
| Forecast                       | Current AQI + 3-day forecast                  |
| History                        | Recent AQI history                            |
| Weather                        | Temperature, humidity, wind, pollutant driver |
| Health tips / exercise         | Practical guidance                            |
| Alerts                         | Hazardous / unhealthy warnings                |
| Explain                        | SHAP explanations                             |
| Leaderboard / ops / monitoring | Model and pipeline health                     |

Prediction flow:

- Load latest feature row for selected city
- Fill future-weather fields from Open-Meteo forecast
- Run the horizon-specific winner model
- Apply prediction interval
- Map AQI to EPA-style category + color
- Return JSON to the AtmoVista frontend

## 13. AtmoVista dashboard (frontend)

Built with React + Vite + TypeScript and deployed on Vercel.

### For you (everyday mode)

- Pakistan live map + city rankings
- Current AQI gauge with category bands
- Weather strip, pollutant insight, 3-day outlook with calendar dates
- Confidence / peak / improvement insights
- Health tips, alerts, exercise advice, smog season panel, city watchlist

### For experts (technical mode)

- Ops strip (storage mode, model, sync)
- Beat-the-baseline panel and pipeline health board
- Model leaderboard and SHAP explanations
- Live accuracy table (forecast vs actual)

## 14. Alerts and health guidance

| Feature         | Behavior                                                        |
|-----------------|-----------------------------------------------------------------|
| Alerts          | Shown for unhealthy/hazardous current or forecast AQI with date |
| Health tips     | Category-based guidance for masking, exposure, sensitive groups |
| Exercise advice | Decision card for outdoor exercise based on near-term AQI       |

## 15. Explainability (SHAP)

Implemented in `src/inference/explain.py` and shown in the expert dashboard.

| View                  | Purpose                                             |
|-----------------------|-----------------------------------------------------|
| City-level importance | Which features generally drive forecasts for a city |
| Local explanation     | Why the latest forecast is high or low right now    |

Typical important drivers include recent AQI/PM lags, rolling AQI statistics, and weather-linked features.

## 16. Automation / CI-CD (GitHub Actions)

The repository is on GitHub. Scheduled workflows keep the system updated.

| Workflow           | Schedule | What it does                               |
|--------------------|----------|--------------------------------------------|
| feature-hourly.yml | Hourly   | Refresh features + reconcile monitoring    |
| training-daily.yml | Daily    | Retrain models + sync to Hopsworks         |
| monitor-daily.yml  | Daily    | Maintain scored forecast-vs-actual history |

Repository secrets used by Actions: `HOPSWORKS_API_KEY`, `HOPSWORKS_PROJECT`, `HOPSWORKS_HOST` (optional).

## 17. Deployment

| Service                | Platform                            | Status               |
|------------------------|-------------------------------------|----------------------|
| AtmoVista frontend     | Vercel                              | Deployed             |
| FastAPI backend        | Render ( <code>render.yaml</code> ) | Deployed             |
| Feature store / models | Hopsworks                           | Live source of truth |

Production configuration:

- Render uses `STORAGE_MODE=hopsworks` and Hopsworks credentials
- Vercel uses `VITE_API_URL` pointing to the Render API
- CORS allows the Vercel frontend origin

## 18. Creative / unique additions

| Addition                        | Why it helps                             |
|---------------------------------|------------------------------------------|
| 5 Pakistan cities               | Broader coverage than a single-city demo |
| Dual audience UI                | Everyday users and technical reviewers   |
| Live weather + pollutant driver | Context around the AQI number            |
| Prediction confidence bands     | Uncertainty communication                |
| Peak / improvement insights     | Actionable short-term planning           |
| Smog season calendar            | Seasonal awareness                       |



| Addition                   | Why it helps                   |
|----------------------------|--------------------------------|
| Exercise recommendation    | Practical health decision      |
| Live monitoring board      | Forecast vs actual trust check |
| Beat-persistence baseline  | Shows model value clearly      |
| City watchlist             | Personalization                |
| Calendar dates on horizons | Clearer forecast timing        |

## 19. Limitations

- Open-Meteo AQI is model-based and may differ from local ground stations.
- Longer horizons (+48h / +72h) have higher error by nature.
- Free-tier Hopsworks / Render / Vercel limits can affect latency and cold starts.
- LSTM was trained and compared, but classical models currently win most horizons.

## 20. Future work

- Fuse station-level AQI (where available) for calibration
- Probabilistic category forecasts (not only point AQI)
- City-specific specialist models
- Push notifications for hazardous alerts
- Longer live monitoring window for stronger trust metrics
- Stronger uptime monitoring for the public demo

## 21. Conclusion

AtmoVista completes the internship requirements for an end-to-end AQI prediction system: data ingestion, feature store, multi-model training, registry, automation, explainability, alerts, and an interactive dashboard. The project is implemented, pushed to GitHub, automated with GitHub Actions, and deployed with Render + Vercel + Hopsworks.

Key numbers:

- Strongest forecasts at +24h (validation R2 about 0.72, category accuracy about 73%)
- Live monitoring over 3500 scored rows: overall MAE about 12.3, category accuracy about 75%

These results support practical short-term planning and health guidance.

## 22. Appendix - important files

| Path                           | Role                         |
|--------------------------------|------------------------------|
| pipelines/feature_pipeline.py  | Hourly feature refresh       |
| pipelines/backfill.py          | Historical training data     |
| pipelines/training_pipeline.py | Model training and selection |
| pipelines/hopsworks_sync.py    | Feature and model sync       |
| pipelines/monitor_pipeline.py  | Forecast vs actual logging   |

| Path                                 | Role                |
|--------------------------------------|---------------------|
| api/main.py                          | FastAPI serving     |
| frontend/src/App.tsx                 | AtmoVista dashboard |
| artifacts/model_leaderboard.json     | Training results    |
| notebooks/eda.ipynb                  | EDA                 |
| .github/workflows/                   | CI/CD automation    |
| docs/AtmoVista_Internship_Report.pdf | This report         |

*Report prepared for internship submission - AtmoVista / Pearls AQI Predictor.*